HISBAH

SMART SHOP SYSTEM

Software Requirements Specification (SRS)

Field	Details
Version	1.1 - Final Technical Review
Status	Approved Development Baseline
Target	20-Day MVP
Primary Platforms	Flutter Mobile + React Web
Backend	PHP / Laravel REST API
Database	MySQL + local Hive storage for offline mode
AI	External AI API Provider
Prepared	23 September 2026
Purpose	Single requirements reference for design, development, testing, and sprint planning

Document Control

Item	Value
Project Name	Hisbah Smart Shop System
Document Type	Software Requirements Specification
Version	1.1
Status	Approved Development Baseline
Reference Standards	IEEE 830-1998 / ISO/IEC/IEEE 29148:2018 (structure and traceability guidance)
Development Constraint	One primary developer with AI-assisted implementation; 20-day MVP
Scope Basis	Small-shop daily operations + AI conversational layer + mobile-first offline workflow
Change Rule	Major requirement changes must be reflected in this SRS before becoming part of the MVP baseline

Document Purpose

This document defines the functional requirements, non-functional requirements, business rules, use cases, data model, system architecture, MVP boundary, and acceptance criteria of Hisbah. It is intended to serve as the single baseline for UI/UX design, Laravel backend implementation, React web development, Flutter mobile development, AI integration, testing, deployment, and subsequent sprint planning.

Baseline principle: The 20-day MVP prioritizes a coherent business flow over feature count. Existing working authentication, store isolation, customer, invoice, debt, payment, and dashboard capabilities should be reused and extended where possible.

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CHAPTER 1 - INTRODUCTION AND STRATEGIC CONTEXT

1.1 Product Overview

Hisbah is a mobile-first smart shop management system for small retail businesses such as grocery stores, mini-markets, neighborhood shops, and other small shops that sell products for cash and/or credit. The system centralizes customers, products, inventory, sales, invoices, debts, payments, and business insights while introducing a conversational AI assistant.

The key interaction model is conversational. A user may describe a business event naturally instead of navigating through multiple forms. For example:

"سجل لابو محمد 3 كراتين مياه، الكرتونة 15، دفع 20 والباقي دين."

The system interprets the request, prepares a structured transaction, validates it against normal backend rules, displays a preview, and executes the transaction only after explicit confirmation. The database and Laravel business rules remain the source of truth.

1.2 Problem Statement

Small-shop owners may still depend on notebooks, memory, phone messages, and manual calculations even when digital POS or accounting products are available. The operational difficulty is not only recording a sale; it is understanding how a sale affects payment, debt, inventory, and the current state of the shop.

- Owners may need to search multiple screens to complete or inspect a simple business operation.
- Credit sales require continuous follow-up on remaining balances, due dates, and overdue accounts.
- Manual inventory tracking can become inaccurate when products are sold quickly or records are delayed.
- Network outages should not stop core local selling and debt-related operations.
- The owner needs actionable information, not only static reports.

1.3 Strategic Objective

Automate and simplify the daily shop cycle while preserving financial and inventory integrity. Hisbah should let the owner operate quickly from mobile, continue core work during temporary connectivity loss, and use natural-language or voice interaction to ask about the business and initiate selected operations safely.

1.4 Target Users

User	Description
Owner	Primary shop decision-maker; manages shop, staff, sales, inventory, customers, debts, and business insights.
Manager	Authorized operational user who can manage approved daily activities and financial operations within assigned permissions.
Employee	Operational user who can perform approved sales/customer/payment functions without unrestricted administrative privileges.
Shop Customer	External recipient of the secure debt/customer statement link.

User	Description
Mentor / Reviewer	Secondary stakeholder reviewing product scope, architecture, and MVP quality.

1.5 Product Objectives

- Provide simple shop management without requiring advanced accounting knowledge.
- Centralize customers, invoices, debts, payments, products, and inventory.
- Maintain correct relationships between sales, payments, debts, returns, and stock.
- Allow core transaction work to continue locally when internet connectivity is unavailable.
- Provide natural-language and voice access to selected business information and actions.
- Require explicit human confirmation before AI-generated write operations are executed.
- Prevent cross-store data access through backend authorization and store scoping.
- Provide both Flutter mobile and React web experiences using the same Laravel API.

1.6 Product Differentiation Hypothesis

The MVP tests the hypothesis that small-shop users may value a simple operational system with natural-language access to real business data more than a system that simply adds more modules. The differentiating experience is therefore not the presence of AI by itself, but the combination of fast shop workflows, offline continuity, and controlled conversational interaction.

```
Natural Language / Voice -> Understanding -> Business Data or Action Draft -> Validation ->
Preview -> User Confirmation -> Business Transaction
```

1.7 MVP Constraints

- One primary developer is responsible for backend, web, mobile, and UI/UX implementation, with AI-assisted development.
- The target implementation window is approximately 20 development days.
- External AI services introduce connectivity, latency, and usage-cost dependencies.
- Core offline transactions depend on local device storage and later synchronization.
- The MVP is not a complete ERP or accounting suite.
- The first mobile release prioritizes Android testing; Flutter keeps the codebase cross-platform where practical.

1.8 MVP Scope

Domain	In Scope
Accounts	Registration, OTP phone verification for store creation, login/logout, roles, invitations, store isolation.
Customers	CRUD, financial summary, statement, debts, payments, secure debt link, WhatsApp share.
Products & Inventory	Product CRUD, barcode, repeated-scan quantity increment, stock adjustments, movements, low-stock alerts.
Sales	Cash, credit, partial payment, invoice items, stock update, returns.
Debts & Payments	Existing debts, due dates, overdue status, partial/full payments, reversals, history.
Dashboard	Daily/weekly/monthly/yearly metrics and actionable alerts.

Domain	In Scope
AI	Business questions, daily summary, invoice/payment draft actions, preview/confirmation, voice input.
Offline	Local storage, FIFO queue, idempotent synchronization, sync state indicator, queue limit.
Web	Administrative/management dashboard and core web workflows.
Mobile	Primary daily operations, barcode, debts, payments, dashboard, AI, voice.

1.9 Out of Scope for MVP

- Advanced supplier and purchase-order management.
- Full accounting/general ledger.
- Multi-branch management.
- Multi-currency customer-facing debt interface.
- Complex tax accounting reports.
- Electronic payment gateway integration.
- Advanced predictive forecasting or machine-learning demand prediction.
- Autonomous AI purchasing or unrestricted database changes.
- Automatic WhatsApp campaigns.
- Full customer portal beyond the secure statement link.
- General-purpose in-app messaging or voice/video calling.
- Advanced warehouse/multi-warehouse management.
- Full OCR invoice processing.

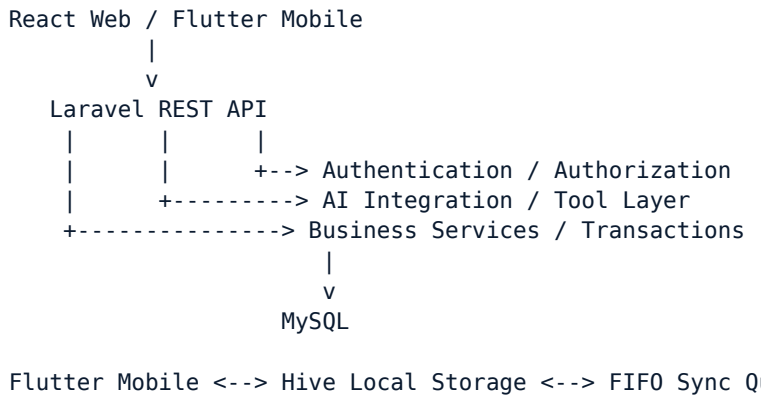
CHAPTER 2 - TECHNICAL ARCHITECTURE AND SYSTEM STRATEGY

2.1 Technology Stack

Layer	Technology	Primary Responsibility
Backend	PHP / Laravel	Business logic, REST APIs, validation, authorization, transactions, AI tool orchestration, synchronization.
Mobile	Flutter / Dart	Fast daily shop operations, barcode scanning, offline-first local workflows, voice input.
Web	React.js	Administrative and management dashboard, data review, configuration, AI interaction.
Database	MySQL	Cloud source of truth for business records.
Local Storage	Hive	Local device storage for offline operational data and synchronization queue.
Design	Figma	Wireframes, design system, UI/UX prototypes.

Layer	Technology	Primary Responsibility
Testing	Postman + automated tests	API verification, regression, integration, core business tests.
Version Control	Git / GitHub	Source control and release workflow.

2.2 System Architecture



2.3 Offline-First Strategy

Hisbah shall support core local operations while the device is offline. Local data is stored through Hive and transactions that cannot immediately reach the cloud are placed into a FIFO synchronization queue. Each queued transaction receives a client transaction identifier so repeated transmission can be detected and made idempotent.

Element	Requirement
Local storage	Core operational data and pending transactions are retained locally while offline.
FIFO queue	Queued transactions are synchronized in the original order in which they were created.
Idempotency	A resent transaction with the same client transaction identifier must not create a duplicate financial entry.
Sync status	UI exposes synchronized, synchronizing, or local-only/pending status.
Queue limit	The local synchronization queue is limited to 1000 pending transactions for MVP; the app should warn and enforce a connection/recovery path before exceeding the limit.
Recovery	Failed synchronization must retain the pending item for retry rather than silently discarding it.

Offline rule: AI interpretation still depends on the external AI service. When the device is offline, the system must prioritize the non-AI local transaction workflows; AI-dependent requests display a clear unavailable state. After a confirmed AI action is created while online, the resulting business transaction follows the same validated local/cloud transaction pipeline.

2.4 Synchronization Semantics

For MVP reliability, a confirmed local transaction is preserved locally until the cloud acknowledges the same client transaction identifier. A synchronization conflict must never delete or silently overwrite a confirmed local financial transaction. The final implementation should expose a recoverable conflict state rather than silently choosing a different business result.

2.5 AI Safety and Control Layer

```
Natural Language / Voice
        -> Intent Recognition
        -> Controlled Tool Selection
        -> Laravel Validation
        -> Action Preview
        -> Explicit User Confirmation
        -> Local or Cloud Transaction
        -> Database / Sync Queue
```

The AI must never execute unrestricted SQL or direct arbitrary model mutations. Read requests are mapped to approved read tools. Write requests are converted into structured drafts and must pass normal backend validation and explicit user confirmation.

2.6 Mobile-First and Barcode Strategy

The mobile POS workflow is optimized for speed and camera-based product identification. Barcode scanning identifies a product and adds it to the cart. Repeatedly scanning the same barcode in the active cart increases the quantity by one instead of creating duplicate product lines.

CHAPTER 3 - SYSTEM SPECIFICATIONS

3.1 Currency

The unified accounting currency of the MVP is the Israeli Shekel (NIS / ₪). Monetary values, invoices, debts, payments, dashboard totals, and AI outputs must use NIS unless a future scope change explicitly introduces additional currencies.

3.2 User Roles and Permissions

Role	Primary Capabilities	Important Restrictions
Owner	All shop administration, staff management, products, inventory, sales, debts, payments, dashboard, AI.	Highest shop-level authority; still subject to system security and validation.
Manager	Customers, products, inventory, sales, payments, debts, dashboard, selected AI functions.	Cannot perform owner-only administration actions unless explicitly authorized.
Employee	Sales, customer lookup, approved payment actions, product lookup, dashboard, selected AI read capabilities.	No unrestricted staff/role management or sensitive financial administration.

3.3 Data Isolation

Every store-owned record must be associated with the correct store. All API queries and AI tools must run within the authenticated store context. A user belonging to Store A must never receive or modify Store B data.

3.4 Customer Secure Debt Link

The system shall generate a non-sequential UUID token for a customer-facing debt statement link. The token must not contain an explicit sequential customer identifier. The link should expire automatically after 30 days and be regenerated when needed. The link can be shared to the customer through WhatsApp using the device/app share interface.

> **Security clarification:** The UUID is used as an opaque, non-sequential public token. The customer record itself is not exposed through a sequential ID in the URL. Authorization and token validation remain server-side responsibilities.

CHAPTER 4 - FUNCTIONAL REQUIREMENTS

ID	Requirement	Description
FR-01	Owner Registration	System shall allow a new owner to create a user account and associated shop.
FR-02	OTP Verification	Creating a new store shall require successful phone OTP verification before activation.
FR-03	Authentication	System shall authenticate registered users and issue an authenticated token/session.
FR-04	Logout	System shall allow the authenticated user to end the current session.
FR-05	Current User	System shall provide current user, role, and shop context.
FR-06	RBAC	Backend shall enforce Owner, Manager, and Employee permissions.
FR-07	Store Isolation	Protected records shall be isolated by store.
FR-08	Staff Invitations	Authorized users shall be able to invite staff to the shop.
FR-09	Staff Management	Owner shall view users, change roles, and remove/deactivate users.
FR-10	Customer Creation	Authorized users shall create customers.
FR-11	Customer Management	Authorized users shall view, update, and deactivate/delete customers subject to rules.
FR-12	Customer Financial Summary	System shall show total debt, paid amount, remaining balance, invoices, and payments.

ID	Requirement	Description
FR-13	Customer Statement	System shall display a chronological statement of relevant invoices, debts, and payments.
FR-14	Customer Debt View	System shall display outstanding debts for a customer.
FR-15	Product Creation	Authorized users shall create products with price, stock, threshold, and optional barcode.
FR-16	Product Management	Authorized users shall view, update, activate/deactivate, and manage products.
FR-17	Barcode Scanning	Mobile POS shall identify products using the phone camera barcode scanner.
FR-18	Repeated Barcode Scan	Repeatedly scanning the same barcode in a cart shall increment quantity by one.
FR-19	Stock Management	System shall maintain current stock for each active product.
FR-20	Stock Adjustment	Authorized users shall increase/decrease stock through controlled adjustments.
FR-21	Stock History	System shall retain stock movement history with reason/source.
FR-22	Low Stock Detection	System shall flag products at or below the configured safety threshold; default MVP threshold is 5 units.
FR-23	Create Invoice	Authorized users shall create invoices containing one or more items.
FR-24	Cash Sale	System shall support fully paid sales.
FR-25	Credit Sale	System shall support unpaid or partially unpaid sales.
FR-26	Partial Payment	System shall support partial payment at invoice creation.
FR-27	Automatic Debt	Unpaid invoice amounts shall create/update the related debt.
FR-28	Inventory Update	Confirmed sales shall update stock as one consistent transaction.
FR-29	Invoice Viewing	Authorized users shall view invoice list and details.
FR-30	Invoice Modification	Invoice changes shall obey financial and stock integrity rules.
FR-31	Invoice Archive/Delete	Invoices linked to immutable historical transactions shall not be hard-deleted; archiving is preferred.

ID	Requirement	Description
FR-32	Sale Return	System shall allow approved sales returns with item-level quantity selection.
FR-33	Return Inventory Update	Returned quantities shall be added back to inventory and related financial balances updated according to return rules.
FR-34	Existing Debt	Authorized users shall record existing/old debts not created by the system.
FR-35	Debt Details	System shall show original, paid, remaining, customer, invoice, creation date, due date, and status.
FR-36	Due Date	A debt may have a due date.
FR-37	Overdue	System shall identify unpaid debts past their due date.
FR-38	Payment Recording	Authorized users shall record full or partial payments.
FR-39	Payment Reversal	Authorized users with sufficient permission shall reverse payments while preserving integrity and auditability.
FR-40	Payment History	System shall show payment history for customer, debt, invoice, and shop contexts.
FR-41	Outstanding Debt	System shall calculate total outstanding debt for the shop.
FR-42	Due/Overdue Lists	System shall support due-today, due-this-week, overdue, and highest-balance lists.
FR-43	Secure Debt Link	System shall generate a non-sequential UUID-based customer debt statement link with 30-day expiry.
FR-44	WhatsApp Share	System shall provide a share action that can open WhatsApp with the secure link and accompanying text.
FR-45	Unified NIS Currency	All MVP accounting values shall use NIS/₪.
FR-46	Dashboard Summary	Dashboard shall show sales, invoices, collections, new debt, outstanding debt, customer count, and low stock.
FR-47	Dashboard Time Filters	Dashboard shall support today, week, month, and year filters.
FR-48	Actionable Insights	Dashboard shall identify due/overdue debts, low stock, and important high-balance conditions.
FR-49	AI Conversation	Authenticated users shall have an AI assistant interface.

ID	Requirement	Description
FR-50	AI Business Questions	AI shall understand supported natural-language questions about authorized shop data.
FR-51	AI Read Tools	MVP shall support at least daily sales, total debt, customer balance, and low-stock retrieval tools.
FR-52	AI Daily Summary	AI shall generate a summary from real shop data for the requested period.
FR-53	AI Action Understanding	AI shall recognize supported invoice and payment actions.
FR-54	AI Structured Draft	AI shall convert supported write requests into structured action drafts.
FR-55	AI Validation	Backend shall validate AI-generated action data using normal business rules.
FR-56	AI Preview	System shall display the exact proposed operation before a write occurs.
FR-57	AI Confirmation	Explicit human confirmation shall be required before execution of AI write actions.
FR-58	AI Invoice Creation	After confirmation, AI-assisted invoice creation shall use the normal invoice transaction path.
FR-59	AI Payment Recording	After confirmation, AI-assisted payment recording shall use the normal payment path.
FR-60	Voice Input	Mobile and approved web clients shall convert voice to text for the AI layer.
FR-61	AI Scope Control	AI shall only access explicitly allowed tools/actions.
FR-62	AI Data Integrity	AI shall not invent customers, products, prices, balances, quantities, or payment amounts.
FR-63	AI Failure Recovery	AI failures shall not create partial financial records and shall provide retry/recovery.
FR-64	Offline Local Transactions	Supported core sales, returns, and payment/debt operations shall be storable locally while offline.
FR-65	FIFO Sync	Pending local transactions shall synchronize in FIFO creation order.
FR-66	Idempotent Sync	Resubmitted transactions with the same client transaction identifier shall not duplicate financial effects.
FR-67	Sync Indicator	UI shall show synchronized, synchronizing, or local-only/pending state.

ID	Requirement	Description
FR-68	Sync Queue Limit	Local pending queue shall support up to 1000 transactions before a forced recovery/connectivity path.
FR-69	Optional Location and Time	Where enabled by the product, local transaction time and approximate location may be recorded for audit/traceability.
FR-70	Critical Audit Logging	System shall record actor and timestamp for critical actions, including AI-confirmed writes, payment reversals, and role changes.

CHAPTER 5 - NON-FUNCTIONAL REQUIREMENTS

ID	Area	Requirement
NFR-01	API Performance	Normal authenticated API requests should generally return within 3 seconds under MVP test load, excluding external AI latency.
NFR-02	Barcode Responsiveness	Barcode recognition and product lookup should complete within 1.5 seconds under normal lighting on supported test devices.
NFR-03	Local Search	Manual local product search should return within 0.5 seconds for up to 10,000 locally stored items under the defined test device.
NFR-04	Offline Readiness	Core approved local transaction workflows shall remain usable without internet connectivity.
NFR-05	Local Commit	A confirmed sale should update the local inventory representation within 200 milliseconds on the supported test device, excluding UI animation.
NFR-06	Sync Time	After connectivity is restored at or above the defined 1 Mbps test condition, queued transactions should be synchronized within 60 seconds when the queue is within normal MVP size and service is available.
NFR-07	Web Statement	Secure-link customer statement should load within 3 seconds on a 4G network under the defined test environment.
NFR-08	Security	Authentication, server-side validation, RBAC, store isolation, HTTPS/TLS in production, and secure error handling are mandatory.

ID	Area	Requirement
NFR-09	UUID Link Security	Customer public debt links shall not expose sequential customer IDs and shall use an expiring, non-sequential token.
NFR-10	Reliability	Critical multi-record financial operations shall be atomic and retry-safe.
NFR-11	Data Integrity	Foreign keys, validation, stock constraints, and trusted server-side calculations shall preserve business consistency.
NFR-12	Usability	Primary mobile workflows should require minimal navigation and clear business language.
NFR-13	Flutter Maintainability	Mobile implementation shall use maintainable architecture and stable dependencies suitable for long-term support.
NFR-14	Accessibility	Core web workflows should provide clear labels, readable text, visible validation, sufficient contrast, and keyboard support where practical.
NFR-15	Privacy	Customer and shop data shall only be exposed to authorized users and authorized secure-link holders.
NFR-16	Backup and Recovery	Production database backups and recovery documentation shall be configured according to the selected hosting environment.

CHAPTER 6 - BUSINESS RULES, PRIORITIES, AND ACCEPTANCE CRITERIA

Code	Rule	Definition
BR-01	Credit Limit	A credit sale shall be rejected when the customer exceeds the configured credit limit, when credit limits are enabled for that customer.
BR-02	AI Confirmation	No AI-proposed financial write operation is executed before explicit user confirmation.
BR-03	Currency	NIS/₪ is the unified accounting currency in the MVP.
BR-04	Product History	A product linked to previous sales should not be hard-deleted; archive/deactivate instead.
BR-05	FIFO	Offline queued financial transactions synchronize in FIFO order.

Code	Rule	Definition
BR-06	Secure Link	Debt statement links use non-sequential UUID tokens and never expose explicit customer IDs in the URL.
BR-07	Closed Invoice	Once a transaction is finalized and synchronized, item quantities are not freely edited; corrections use controlled return/adjustment workflows.
BR-08	Time / Location	Local transaction time is retained; approximate location is optional and only captured when the feature is enabled and permitted.
BR-09	OTP	New store activation requires successful phone OTP verification.
BR-10	Conflict Handling	A confirmed local financial transaction must not be discarded during synchronization; unresolved conflicts enter a recoverable conflict state.
BR-11	Low Stock	Default MVP safety threshold is 5 units unless a product has a different configured minimum threshold.
BR-12	Queue Limit	Local pending synchronization should not exceed 1000 items without a recovery/connectivity action.
BR-13	AI Authorization	AI cannot grant permissions that the authenticated user does not already have.
BR-14	AI Source of Truth	Database and backend business rules are the source of truth; AI output is interpretation or controlled decision-support.

6.1 MoSCoW Prioritization

Priority	MVP Items
Must Have	Authentication + OTP, store isolation, customers, products, barcode, inventory, sales, returns, debts, payments, NIS, offline local transactions, FIFO sync, idempotency, dashboard basics, AI read tools, AI invoice/payment drafts, confirmation, mobile/web core flows, testing, deployment.
Should Have	WhatsApp secure debt-link sharing, stronger actionable insight cards, customer statement refinements, stock movement history UI, richer AI daily summary.
Could Have	Bluetooth invoice printing, advanced visual analytics, additional AI read tools beyond the approved minimum, deeper theme customization.
Won't Have	Full accounting/GL, multi-currency interfaces, electronic payment gateway, multi-branch ERP, autonomous AI, advanced forecasting, general messaging/video.

6.2 Acceptance Criteria

#	Acceptance Criterion	Verification
AC-01	A barcode scan identifies a product and adds it to the cart within the defined performance target.	System / Device Test
AC-02	Repeating the same barcode scan increments the cart quantity rather than creating a duplicate line.	Functional Test
AC-03	A credit invoice correctly updates invoice total, payment, debt, and stock.	Integration Test
AC-04	A returned quantity is restored to stock and the related balance is recalculated according to the return rule.	Integration Test
AC-05	A generated customer link contains no sequential customer ID and expires after 30 days.	Security / Validation Test
AC-06	During a network outage, an approved offline transaction is stored locally and later synchronized exactly once.	Subsystem Verification
AC-07	Ten queued transactions synchronize in FIFO order after reconnection without duplicates.	Subsystem Verification
AC-08	An AI invoice request displays a complete preview and executes only after explicit confirmation.	AI Integration Test
AC-09	An AI tool cannot retrieve data from another store.	Security Test
AC-10	The dashboard displays sales, collections, debt, and low-stock metrics for the selected period.	System Test

CHAPTER 7 - USE CASE SPECIFICATION

ID	Use Case	Primary Actor
UC-01	Register Owner and Store	Owner
UC-02	Verify Store Registration by OTP	Owner
UC-03	Login / Logout	Owner / Manager / Employee
UC-04	Manage Staff and Roles	Owner
UC-05	Manage Customers	Owner / Manager / Employee
UC-06	View Customer Statement	Authorized User / Customer via secure link
UC-07	Manage Products	Owner / Manager
UC-08	Scan Product by Barcode	Owner / Manager / Employee

ID	Use Case	Primary Actor
UC-09	Manage Inventory	Owner / Manager / authorized Employee
UC-10	Create Cash Sale	Owner / Manager / Employee
UC-11	Create Credit or Partial Sale	Owner / Manager / Employee
UC-12	Process Sale Return	Owner / Manager
UC-13	Record Payment	Authorized User
UC-14	Reverse Payment	Owner / Manager
UC-15	Manage Debts	Owner / Manager
UC-16	Generate and Share Secure Debt Link	Owner / Manager
UC-17	View Dashboard and Alerts	Authorized User
UC-18	Ask AI Business Question	Authorized User
UC-19	Create Invoice Using AI	Authorized User
UC-20	Record Payment Using AI	Authorized User
UC-21	Use Voice Input	Authorized User
UC-22	Synchronize Offline Transactions	System / Authorized User monitors status

UC-01 - Register Owner and Store

Primary Actor: Owner

Preconditions: User is not authenticated; registration service is available.

Main Flow:

- Open registration
- Enter owner and shop information
- Verify phone using OTP
- System validates data
- System creates user and store
- System assigns Owner role
- System returns authenticated or successful registration state

Exceptions: Invalid data -> validation error | Duplicate account -> reject | OTP failure/expiry -> retry

Postcondition: Owner and store exist with verified registration state.

UC-02 - Verify Store Registration by OTP

Primary Actor: Owner

Preconditions: Owner has submitted a valid phone number.

Main Flow:

- System sends OTP
- Owner enters code

- System validates code and expiry
- System marks phone verified
- Registration flow proceeds

Exceptions: Invalid/expired code -> reject and allow resend | Service unavailable -> show retry

Postcondition: Phone verification is recorded.

UC-08 - Scan Product by Barcode

Primary Actor: Owner / Manager / Employee

Preconditions: Sales screen is open; camera permission is available.

Main Flow:

- Activate scanner
- Capture barcode
- System finds product in local/cloud catalog
- Add product to cart
- If same product exists, increment quantity

Exceptions: Barcode not found -> show clear error | Camera unavailable -> allow manual search

Postcondition: Product is present in cart with correct quantity.

UC-10 - Create Cash Sale

Primary Actor: Owner / Manager / Employee

Preconditions: Products exist and sufficient stock is available.

Main Flow:

- Select/scan products
- Review quantities
- System calculates total
- Receive full payment
- Confirm sale
- Commit invoice, payment, and stock movement

Exceptions: Insufficient stock -> reject | Validation failure -> do not commit

Postcondition: Cash invoice is recorded and stock is updated.

UC-11 - Create Credit or Partial Sale

Primary Actor: Owner / Manager / Employee

Preconditions: Customer exists.

Main Flow:

- Select customer
- Add products
- Enter paid amount
- System calculates remaining amount
- Apply credit-limit rule when enabled

- Confirm
- Commit invoice, payment, debt, and stock atomically

Exceptions: Credit limit exceeded -> reject | Insufficient stock -> reject

Postcondition: Invoice and related debt/payment/stock records are consistent.

UC-12 - Process Sale Return

Primary Actor: Owner / Manager

Preconditions: Original sale exists and return is permitted.

Main Flow:

- Open original invoice
- Select returned items/quantities
- System validates returnable quantity
- Calculate return amount
- Confirm return
- Restore stock and adjust financial relationship

Exceptions: Returned quantity exceeds sold quantity -> reject | Closed/synchronized rule violation -> use controlled adjustment

Postcondition: Return is recorded and inventory/balance are updated consistently.

UC-16 - Generate and Share Secure Debt Link

Primary Actor: Owner / Manager

Preconditions: Customer has financial record.

Main Flow:

- Open customer
- Request secure link
- System generates opaque UUID token
- System applies 30-day expiry
- User selects share/WhatsApp
- System opens share flow with link and text

Exceptions: Link generation failure -> retry | Expired token -> regenerate

Postcondition: Customer can access the time-limited statement link.

UC-18 - Ask AI Business Question

Primary Actor: Authorized User

Preconditions: User is authenticated; AI service is reachable.

Main Flow:

- Open assistant
- Send text or voice request
- AI identifies supported intent
- Backend executes approved read tool
- AI receives trusted data

- Assistant returns response

Exceptions: AI unavailable -> recoverable error | Unsupported request -> clear limitation message

Postcondition: User receives a response based on authorized real shop data.

UC-19 - Create Invoice Using AI

Primary Actor: Authorized User

Preconditions: User has invoice authority; AI service is reachable.

Main Flow:

- Send natural-language text or voice request
- AI identifies customer/products/amounts
- Backend validates entities and rules
- System creates draft
- UI displays preview
- User confirms
- Normal invoice transaction executes

Exceptions: Unknown product/customer | Ambiguous request | Invalid quantity/price | Insufficient stock | User cancels | AI failure

Postcondition: Confirmed invoice, payment, debt, and stock changes are committed exactly once.

UC-20 - Record Payment Using AI

Primary Actor: Authorized User

Preconditions: User has payment authority.

Main Flow:

- Send request such as "أبو محمد دفع 50 شيكل"
- AI identifies customer and amount
- Backend validates outstanding balance
- Preview displayed
- User confirms
- Payment is recorded

Exceptions: Unknown customer | Invalid amount | AI failure | User cancels

Postcondition: Payment is recorded and remaining debt is updated.

UC-22 - Synchronize Offline Transactions

Primary Actor: System / Authorized User monitors status

Preconditions: Device has queued transactions and connectivity is available.

Main Flow:

- Load oldest queued transaction
- Send with client transaction identifier
- Backend validates and checks idempotency

- Commit acknowledged transaction
- Mark local item synchronized
- Continue FIFO until queue is empty

Exceptions: Network failure -> retain item for retry | Duplicate client transaction -> no duplicate effect | Conflict -> recoverable conflict state

Postcondition: Cloud and local records are reconciled without duplicate financial effects.

CHAPTER 8 - DATA MODEL AND ERD

8.1 ERD Overview

The data model extends the existing financial core with products, inventory, sale returns, secure customer links, OTP verification, AI conversations/action drafts, and a local synchronization queue. The visual ERD in Appendix B distinguishes cloud/MySQL entities from the local Hive queue.

Entity	Purpose	Main Logical Attributes
Store	Shop/business record	id, name, phone, address, status
User	Authenticated system user	id, store_id, name, email, role, status
Customer	Shop customer	id, store_id, name, phone, notes, status
Product	Sellable product	id, store_id, name, barcode, selling_price, cost_price, current_stock, minimum_stock, status
StockMovement	Product quantity change	id, store_id, product_id, type, quantity, reason/source, reference, created_by
Invoice	Sales transaction	id, store_id, customer_id, totals, type, status, created_by
InvoiceItem	Product line in invoice	id, invoice_id, product_id, quantity, unit_price, line_total
Debt	Outstanding customer amount	id, store_id, customer_id, invoice_id, original/paid/remaining, due_date, status
Payment	Customer payment	id, store_id, customer_id, payment_group_id, amount, payment_date, status, created_by
PaymentGroup	Grouped payment/reversal context	id + group metadata
SaleReturn	Return transaction	id, store_id, invoice_id, customer_id, total_amount, reason, status, created_by
SaleReturnItem	Returned item	id, return_id, invoice_item_id, product_id, quantity, amount
CustomerDebtLink	Public customer statement token	id, customer_id, token, expires_at, status
AIConversation	User AI session	id, store_id, user_id, status, started_at, ended_at

Entity	Purpose	Main Logical Attributes
AIMessage	AI conversation message	id, conversation_id, sender, content, type, sequence, timestamp
AIActionDraft	AI write-action proposal	id, conversation_id, user_id, action_type, payload, validation_status, confirmation_status, executed_at
AuditLog	Critical action record	id, store_id, user_id, action, entity_type/id, metadata, timestamp
OTPVerification	Phone verification record	id, user/phone reference, code/status, expires_at, verified_at
OfflineSyncQueue	Local/Hive queue item	local_id, client_transaction_id, type, payload, sequence, status, retry_count, created_at

8.2 Core Relationships

Entity A	Cardinality	Entity B
Store	1 : *	User
Store	1 : *	Customer
Store	1 : *	Product
Store	1 : *	Invoice
Store	1 : *	Debt
Store	1 : *	Payment
Product	1 : *	StockMovement
Invoice	1 : *	InvoiceItem
Product	1 : *	InvoiceItem
Customer	1 : *	Invoice
Customer	1 : *	Debt
Customer	1 : *	Payment
Invoice	0..1 : 1	Debt
Debt	1 : *	Payment
PaymentGroup	1 : *	Payment
Invoice	1 : *	SaleReturn
SaleReturn	1 : *	SaleReturnItem
InvoiceItem	1 : *	SaleReturnItem
Product	1 : *	SaleReturnItem
Customer	1 : *	CustomerDebtLink
User	1 : *	AIConversation
AIConversation	1 : *	AIMessage
AIConversation	1 : *	AIActionDraft
User	1 : *	AuditLog

Entity A	Cardinality	Entity B
User	1 : *	OTPVerification

8.3 Data Integrity Rules

- Every store-owned entity is scoped to its store.
- Invoice items must reference products belonging to the same store as the invoice.
- Debts and payments must reference customers and financial records within the authorized store.
- Stock movement records must explain quantity changes.
- Financial operations affecting invoice, payment, debt, and stock must be atomic where the workflow requires multiple records to change.
- AI tools must execute within the authenticated store context.
- AI action payloads must be validated by Laravel before execution.
- Offline queued operations must carry a unique client transaction identifier and preserve FIFO order.
- Customer debt links must be non-sequential tokens with explicit expiry.
- Critical actions retain actor/timestamp information.

CHAPTER 9 - PLATFORM RESPONSIBILITIES

9.1 Backend Responsibilities

- Authentication and OTP verification
- Authorization and store isolation
- Customers and customer financial records
- Products and inventory
- Invoices and returns
- Debts and payments
- Dashboard calculations
- AI tool orchestration and structured action validation
- Offline synchronization/idempotency
- Audit logging
- REST API documentation and error handling

9.2 Flutter Mobile Responsibilities

- Fast POS workflow
- Camera barcode scanning
- Local Hive data
- Offline transaction queue
- Sync status indicator
- Customers/debts/payments
- Inventory overview
- Dashboard
- AI assistant
- Voice input
- Primary daily operational flows

9.3 React Web Responsibilities

- Dashboard and management views

- Customers and statements
- Products and inventory
- Invoices, returns, debts, and payments
- Staff and role management
- AI assistant and action confirmation
- Secure-link statement web view
- Responsive review/administration experience

9.4 UI/UX Responsibilities

- Mobile-first daily flow design
- Fast POS and barcode interaction
- Clear paid/unpaid/due/overdue states
- Visible low-stock states
- Offline/online synchronization status
- Explicit AI preview and confirmation states
- Readable Arabic/business terminology where applicable
- Minimal navigation and clear error/empty/success states

CHAPTER 10 - MVP ACCEPTANCE BOUNDARY

Area	MVP Acceptance Condition
Accounts	Owner can create a store after OTP verification; users can authenticate with correct roles.
Store Isolation	No user or AI tool can access data belonging to another store.
Customers	Authorized users can manage customers and view financial statements.
Products	Authorized users can manage products and configure stock thresholds/barcodes.
Inventory	Sales and returns update stock consistently.
POS	Barcode scan and manual product selection support cash/credit/partial sale flows.
Debts	Existing and invoice-generated debts can be tracked with due/overdue state.
Payments	Partial/full payments and authorized reversals work correctly.
Secure Link	Customer can open an expiring statement link without a sequential customer ID in the URL.
Offline	Core approved local operations work without internet and are later synchronized.
Sync	FIFO and idempotency prevent ordering errors and duplicates.
Dashboard	Sales, collections, debts, customers, and low-stock metrics are available.
AI Read	Supported business questions use real authorized database data.
AI Action	Invoice/payment actions produce previews and require explicit confirmation.

Area	MVP Acceptance Condition
Voice	Voice input feeds the same AI logic as text.
Web	Core management workflows are functional in React.
Mobile	Primary daily workflows are functional in Flutter.
Quality	Core security, integrity, offline, and financial integration tests pass.
Deployment	The approved MVP is deployable and recoverable in the selected environment.

CHAPTER 11 - TESTING AND QUALITY BASELINE

11.1 Backend Testing

- Authentication and OTP
- Store isolation
- Role authorization
- Customer CRUD
- Product CRUD
- Barcode lookup behavior
- Invoice creation
- Credit/partial sale
- Sale return
- Debt creation/update
- Payment and reversal
- Stock update
- Secure link expiry
- AI authorization
- AI action validation
- Offline synchronization and idempotency

11.2 Integration Tests

```
Flow A
Create Product -> Create Customer -> Create Invoice -> Payment -> Debt -> Stock

Flow B
Offline Sale -> Local Storage -> FIFO Queue -> Reconnect -> Idempotent Sync -> Cloud
Acknowledgment

Flow C
AI Request -> Intent -> Tool / Draft -> Validation -> Preview -> Confirm -> Business
Transaction
```

11.3 V-Model Verification Mapping

Requirement Type	Verification
Functional requirements	Unit + integration + system testing
Performance requirements	Performance/device/network testing
Security requirements	Authorization, isolation, token/link security testing

Requirement Type	Verification
Offline requirements	Subsystem synchronization tests
AI requirements	AI integration and adversarial/validation tests
Usability requirements	Task-based usability checks on mobile/web
Acceptance boundary	End-to-end MVP demonstration scenarios

CHAPTER 12 - REQUIREMENTS TRACEABILITY AND CHANGE CONTROL

Every implementation artifact should reference this SRS. Sprint planning will be created only after the SRS baseline is accepted.

Artifact	Required Reference
Epic	Business capability / MVP module
User Story	Relevant FR ID(s) + UC ID
Acceptance Criteria	Observable behavior derived from FR/NFR
Backend Task	FR / UC implemented
Flutter Task	FR / UC implemented
React Task	FR / UC implemented
UI/UX Task	FR / UC being designed
AI Task	FR-49 to FR-63 + relevant UC
Offline Task	FR-64 to FR-68 + related NFRs
Test Case	FR/NFR + UC verified
Bug	Affected FR/UC/NFR
Change Request	Existing requirement + proposed change + affected model/use case

12.1 Baseline Decision

Final MVP principle: The minimum coherent product is not "an ERP with an AI chatbot". It is a small-shop operating system that combines daily sales, customer/debt management, inventory, offline continuity, and a controlled conversational interface for business information and selected transactions.

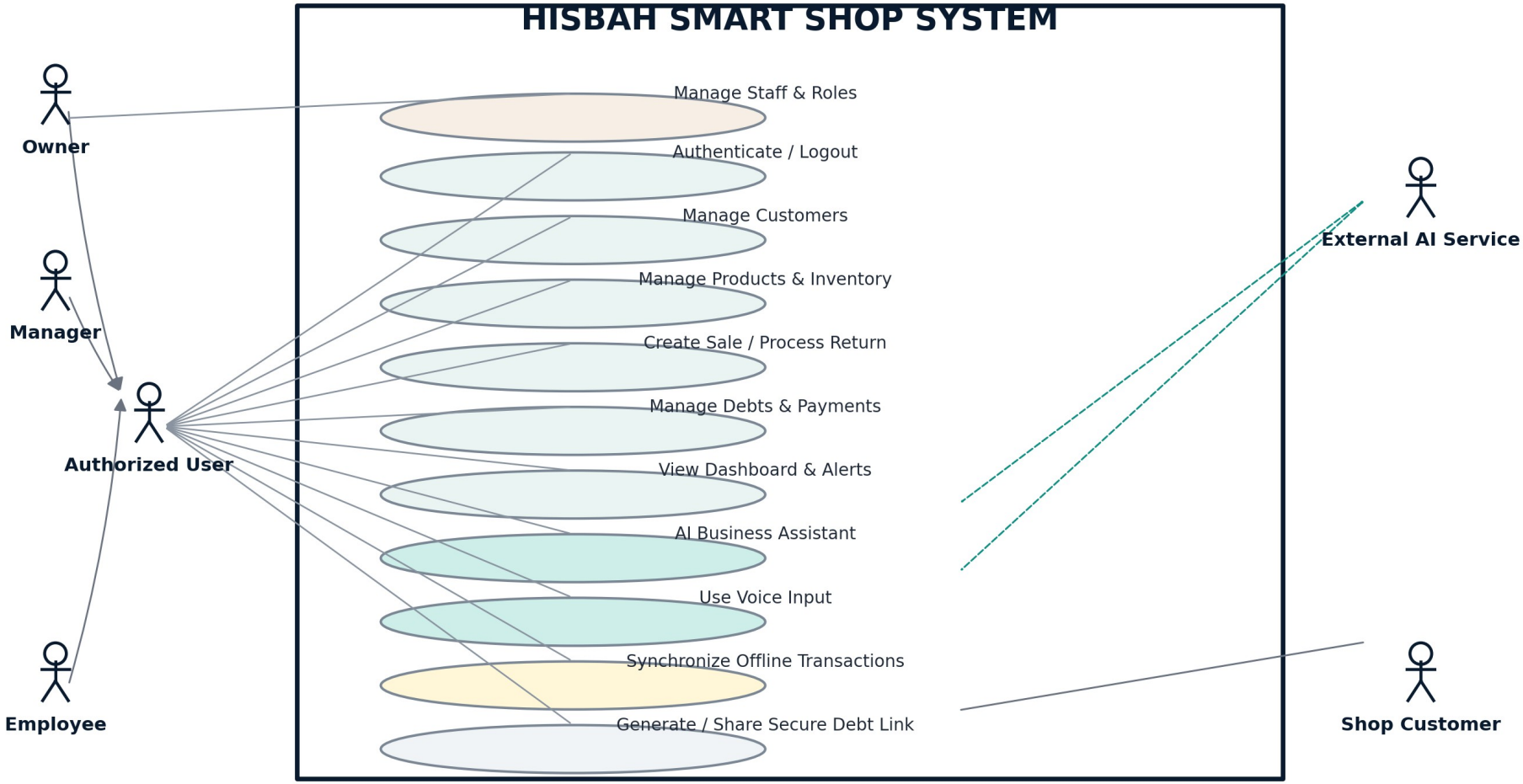
12.2 Items Intentionally Deferred

- Advanced supplier/purchase management
- Full accounting/GL
- Multi-branch
- Multi-currency
- Electronic payments
- Advanced forecasting
- Autonomous AI

- Automatic messaging campaigns
- Full customer portal
- General messaging/video
- Full OCR
- Advanced warehouse management

APPENDIX A - VISUAL UML USE CASE DIAGRAM

Figure A-1. Visual Use Case Diagram for Hisbah Smart Shop.

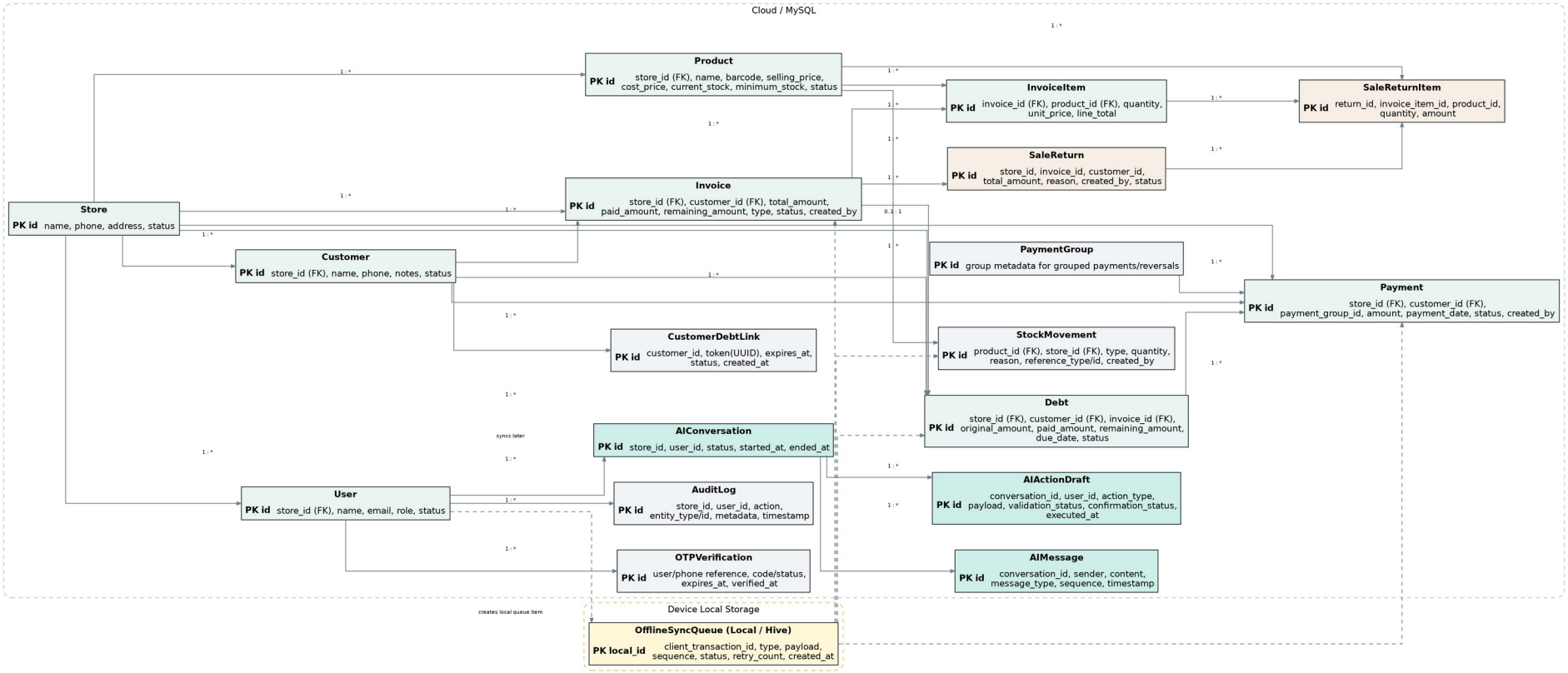


Overview diagram: detailed requirements and individual use cases are specified in Chapter 7 and the requirements catalogue.

The diagram focuses on primary actors and the core MVP interactions. Synchronization is shown as a system-level behavior monitored by authorized users; the external AI service participates in AI question/action flows.

APPENDIX B - VISUAL UML ERD

Figure B-1. Visual ERD for the Hisbah MVP. Cloud/MySQL entities are grouped separately from the local Hive synchronization queue.



The ERD includes the existing financial core plus inventory, returns, secure customer links, OTP verification, AI interaction entities, audit logging, and the local offline synchronization queue.